

Analysis of Inversion Attack and Defense Strategies on Speaker Classification Models

Xuan-Yu Lin¹, Chia-Wei Chuang²

^{1,2}National Hualien Senior High School, Hualien, Taiwan.

ABSTRACT

Recent advances in Deepfake voice generation have intensified concerns over speech privacy, as creating realistic synthetic voices requires access to representative speaker features. Model inversion attacks make this feasible by reconstructing such features directly from trained recognition models. However, prior studies rarely examined how model architecture itself governs this vulnerability. This work bridges that gap by analyzing the architectural determinants of inversion susceptibility and identifying when differential privacy offers meaningful defense. From the attacker's perspective, robustness arises from waveform-based architectures with sufficient depth — the Raw model provides the strongest protection, while SincNet balances convergence and defense through learnable filters. In contrast, MFCC-based models discard critical cues early, making inversion easier. We further uncover a defensive “sweet spot” at around 1.6–2.0s input duration. Differential privacy strengthens already robust architectures but fails to rescue weak ones. These findings establish an architecture-level foundation for privacy-preserving speech AI.

Keywords:

Model Inversion Attack
Speaker Recognition
Model Security

Corresponding Author:

Xuan-Yu Lin,
National Hualien Senior High School,
Hualien, Taiwan.
Email: s210347@hlhs.hlc.edu.tw

1. INTRODUCTION

Deepfake voice scams pose a growing threat to digital security, enabling fraud and impersonation through synthetic voices that mimic real individuals. These risks are especially critical for speaker recognition systems, which rely on learned vocal patterns for authentication. A particularly severe vulnerability occurs when model inversion attacks exploit these systems to reconstruct a speaker's voice traits — such as timbre, rhythm, and accent — from a trained model, which can then be abused to generate convincing deepfakes.

Motivated by this concern, our research aims to analyze and mitigate the risks posed by model inversion attacks on speaker recognition systems. Specifically, we investigate how different model architectures affect vulnerability and under what conditions differential privacy can provide effective protection. By understanding these factors, we seek to establish design principles for more secure speaker recognition models and contribute to the development of robust defense strategies.

Our findings show several key insights into building secure speaker recognition systems. We observe that architecture plays a crucial role: end-to-end waveform models demonstrate stronger robustness, with SincNet representing a balanced trade-off between model size, training difficulty, and resilience, while Raw models exhibit the highest resistance to inversion. In terms of model capacity and input length, deeper architectures with audio segments between 1.6 and 2 seconds

achieve the best trade-off between performance and security. Regarding representation, avoiding spectrogram-based preprocessing enhances resistance to inversion attacks. Finally, differential privacy proves most effective in Raw and SincNet architectures, though it offers limited benefits for weaker MFCC baselines. In essence, DP should be viewed as an amplifier of inherent robustness rather than a universal shield against inversion attacks. Collectively, these findings help to design more secure and privacy-resilient speaker recognition models.

1.1. Background

1) Feature extraction

We selected three distinct feature-extraction approaches—MFCC, SincNet, and a Raw-waveform convolutional baseline—explicitly to define the target models for our inversion experiments and to probe how different representations affect privacy leakage. MFCC represents the classical, hand-engineered signal-processing pipeline that compresses speech into a low-dimensional spectral summary reflecting human auditory perception. Because it discards large amounts of fine temporal structure, it serves as a baseline for how compact, information-lossy preprocessing impacts inversion risk. SincNet is an end-to-end architecture that still embeds prior signal knowledge: its first layer uses parameterized sinc filters that learn low/high cutoff frequencies, which speeds convergence and improves interpretability while retaining more waveform detail than MFCC. The Raw model applies standard convolutional filters directly to the waveform with no pre-filtering, providing a maximum-capacity, high-fidelity reference that preserves the richest low-level structure. By treating these three as the attack targets, we can compare how early feature compression, learned filter priors, and full waveform expressiveness trade off between classification utility and vulnerability to gradient-based reconstruction.

2) Attack types

There are several attack classes relevant to model privacy—adversarial attacks, membership inference attacks, and model inversion attacks—but they differ in goals and information requirements. We focus our evaluation on model inversion because it directly reconstructs training-like inputs from model internals. Adversarial attacks seek small perturbations that force misclassification and therefore probe model decision boundaries, while membership inference attempts to determine whether a particular sample was present in training data; both leak different kinds of information but do not aim to reconstruct the original signal. Model inversion instead iteratively optimizes synthetic inputs (often using gradients) so that the target model’s outputs match a selected label or embedding; this produces concrete, human-interpretable reconstructions and therefore represents a severe privacy threat. We run experiments in a white-box setting — where full parameter and gradient access is available — because white-box attacks set an upper bound on attack effectiveness and allow controlled, repeatable comparisons across architectures; black-box reconstruction is substantially harder and less standardized, so it is outside this study’s primary scope.

3) Differential Privacy as Defense

Differential Privacy (DP) is a formal framework designed to protect individual data points by ensuring that small changes in the training dataset do not significantly affect the model’s outputs. In deep learning, DP is commonly implemented by adding noise to training gradients—a process that constrains how much information each data sample can contribute to model updates. While effective in principle, this gradient-based approach is computationally demanding and often destabilizes convergence, especially for high-dimensional data such as speech signals.

To address these challenges, our study adopts a simplified, post-training DP strategy inspired by the One Parameter Defense framework [3]. Instead of modifying the gradient updates during training, this method attaches a lightweight DP layer to the model’s output to achieve privacy

protection more efficiently. This approach preserves the core idea of DP while substantially reducing training cost and convergence issues, making it better suited for speech recognition models used in our experiments.

In summary, this section presents the essential components of our experimental framework. By integrating three feature-extraction approaches, a white-box gradient-based inversion attack, and a lightweight DP defense, we establish a comprehensive setup for evaluation.

1.2. Related Work

1) Model Inversion Attack

In “*Introducing Model Inversion Attacks on Automatic Speaker Recognition*” (Pizzi et al., 2023) [2], the authors proposed a new inversion strategy and compared it with the traditional method on speaker recognition models. They evaluated attack success by passing reconstructed audio through SincNet for feature extraction and a neural classifier for recognition. If the system identified the generated audio as the target speaker, the attack was considered successful; the success rate was measured by the percentage of correctly classified reconstructed samples.

- **Traditional Model Inversion Attack:** The attacker leverages the model’s prediction confidence to reverse-engineer data associated with a specific target. Suppose the target model f maps an n -dimensional input x to an m -dimensional probability vector p . A cost function $c(x) = 1 - p(t)$ is defined for a target class t . Starting from a random initialization x_0 , the attacker applies gradient descent iteratively to minimize $c(x)$. Each iteration updates the input so that the model increasingly classifies it as t . Through this process, the reconstructed data captures class-representative characteristics—such as frequency distribution and vocal texture—resulting in an approximation of the target’s voice.
- **Sliding Model Inversion Attack:** Pizzi et al. further proposed a Sliding Model Inversion Attack, which divides audio into overlapping time windows and inverts each segment sequentially. An inversion vector of length ℓ is initialized, and a fixed-length window slides across it with a set step size. Each windowed section is inverted using gradient updates and merged back into the main vector. The process continues until the loss $c(x)$ reaches a predefined threshold or the maximum iteration count is reached. This localized optimization better preserves temporal consistency, producing smoother and more realistic reconstructions.

The study found that the Sliding Model Inversion Attack significantly improved reconstruction accuracy: the classification accuracy of inverted audio samples increased from 54% (traditional) to 90% (sliding). This result clearly demonstrates the high vulnerability of existing speaker recognition systems when exposed to gradient-level access and highlights the importance of robust defenses—an objective that is central to our research. However, while this study reported that the Sliding Model Inversion Attack could substantially enhance reconstruction accuracy compared to the traditional approach, our findings do not fully align with this observation. In our experiments, the sliding method did not consistently outperform the Full-Gradient one, suggesting that its effectiveness is highly conditional on factors such as model architecture and input configuration, a key point of investigation in our work. It is also worth noting that both the traditional and sliding attack implementations in our experiments were reproduced based on the algorithmic descriptions from the cited study. Since no public implementation (e.g., GitHub repository) was provided, we reconstructed the methods according to their procedural details, which may partially account for the performance discrepancies observed between our results and the original report.

2) Differential Privacy Defense

In our earlier experiments, we adopted a differential privacy (DP) strategy based on the TensorFlow Privacy library. This gradient-based DP algorithm introduces noise into the training gradients to increase the difficulty of data inference attacks. However, we observed that this approach considerably extended the training time of speech models and often caused convergence instability.

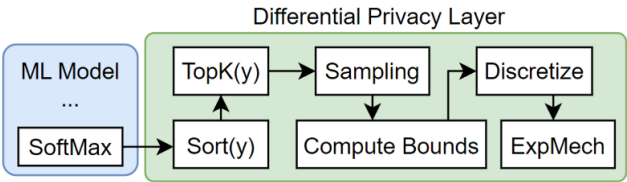


Figure 1: Differential Privacy Algorithm Flowchart

To overcome these issues, we referred to *One Parameter Defense – Defending against Data Inference Attacks via Differential Privacy* [3], which proposed a simplified DP algorithm for image classification systems. Inspired by this work, we streamlined its mechanism and developed a DP-based defense tailored for speech models, as shown in Figure 3–5. The proposed method eliminates the need for model retraining by appending a DP layer to the original model, achieving effective defense while substantially reducing computational and training costs.

Overall, prior research has established strong baselines for both model inversion attacks and their corresponding defenses. Building on these foundations, our work extends previous efforts by systematically analyzing how architectural design, model depth, and input length influence vulnerability, and by evaluating how differential privacy interacts with these factors. Through this analysis, we aim to provide a clearer understanding in speaker recognition models.

2. RESEARCH METHOD

2.1. Methology Overview

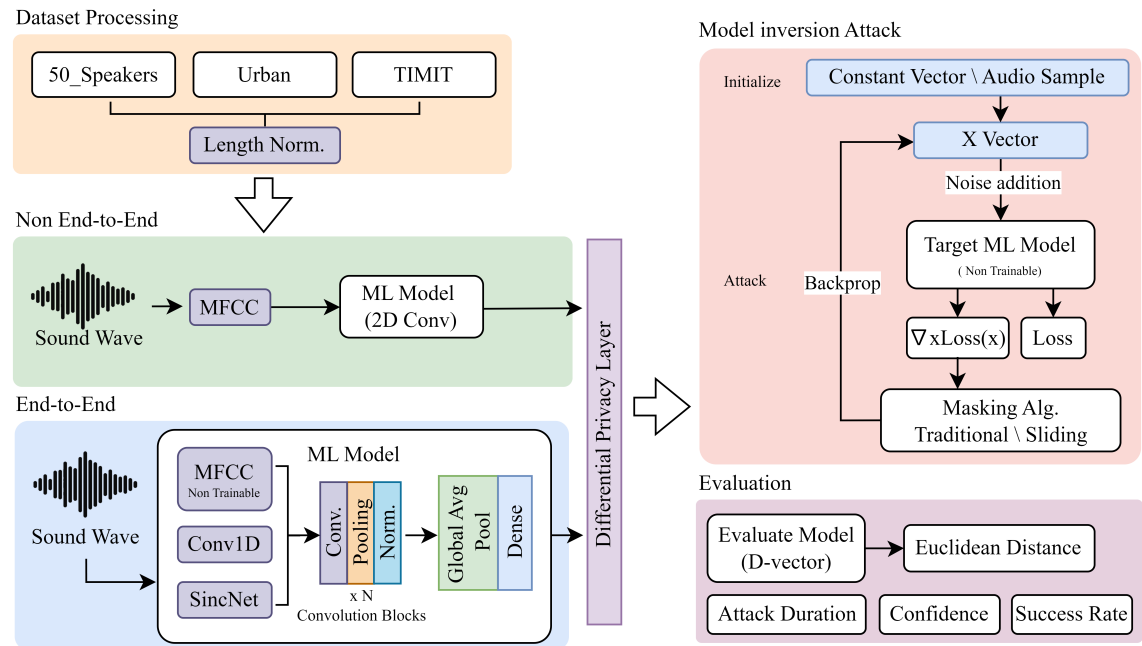


Figure 2: Experiment Design Overview

Our methodology, as illustrated in Figure 2, consists of three main stages: dataset processing, target model training, and model inversion attack with evaluation. In the dataset processing stage, we utilize several speech datasets and apply length normalization to ensure consistent input size. Different normalization strategies are also compared to analyze their effects on the attack performance. Next, we train two types of target models — end-to-end and non-end-to-end. The end-to-end models operate directly on raw waveforms. The non-end-to-end models, on the other hand, use MFCC features as inputs to 2D convolutional networks. Finally, we perform model inversion attacks on the trained classifiers to recover representative audio samples through model’s gradient information. This stage aims to evaluate the susceptibility of different model architectures and privacy settings to inversion-based reconstruction. In this framework, we systematically control five key experimental variables to analyze their impact on model privacy.

Table1: Variables in Experiment

Controlled stage	variables
Dataset processing	Sampling length
Target Model Training	Architecture & input feature
	Model complexity (channel scaling / depth)
	One-Parameter Defense applied or not (binary)
Model Inversion Attack	Seed initialization for inversion (real audio seed vs synthetic seeds)

2.2. Dataset Processing

In this stage, we prepare and standardize audio data for our experiments. All audio recordings are resampled to a fixed rate of 16 kHz, and three public datasets are utilized to ensure the generality of our findings: **50 Speakers**, **UrbanSound**, and **TIMIT**. The 50 Speakers and TIMIT datasets primarily contain human voice recordings, whereas UrbanSound consists mainly of non-speech acoustic events. Combining these sources allows us to examine model behavior under both vocal and environmental sound conditions, thereby enhancing the generalizability of our results. A primary factor controlled during this stage is **Sampling Length**. To examine how input duration influences inversion success and reconstruction quality, we test a range of discrete segment lengths—3200, 6400, 12800, 19200, 25600, 32000, 38400, 44800, and 51200 samples. Each audio file is segmented using a sliding-window approach, and a segment is retained only if more than 50% of its duration contains speech-active content (i.e., non-silent and semantically meaningful acoustic information). This ensures that the extracted samples are informative and avoids training or evaluating on fragments containing mostly silence. For each target length, segments are trimmed or zero-padded to meet the required duration. While a dedicated experiment sweeps across all these lengths, our primary working length for other experiments is 3200 samples (≈ 0.2 s), whereas model-complexity experiments adopt 64000 samples as the standard.

2.3. Target Model Training

We introduce three key variables related to the model's design and defense. First, we evaluate how different model families and input representations affect vulnerability by varying the **Architecture & Feature Type**. This variable compares two main approaches: End-to-end (E2E) models that operate directly on raw waveform inputs, including architectures like RAW and SincNet, as well as an E2E variant where an untrainable MFCC layer converts audio into features for subsequent 1D convolutions. This is contrasted with a Non-End-to-end (Non-E2E) model that uses pre-computed Mel-Frequency Cepstral Coefficient (MFCC) features as input to a 2D convolutional network. By

comparing these, we investigate how the choice of input feature and architecture impacts susceptibility to inversion.

Before discussing **Model Complexity**, it is important to note that all models in our experiments share a common backbone design consisting of four convolutional blocks, each composed of a Convolution → Pooling → Normalization sequence. This baseline configuration is referred to as Structure 0, representing the standard complexity level adopted by most models in our study. To measure whether deeper or wider models are harder to invert, we systematically vary Model Complexity by extending this baseline structure. Specifically, the number of convolutional blocks is incrementally increased—from Structure 0 (4 blocks) up to Structure 4 (8 blocks)—where each additional structure level adds one extra convolutional block to the model. Model complexity is therefore adjusted during model configuration by increasing depth (number of blocks) and by scaling convolutional channel width.

In our results, we analyze the relationship between reconstruction error and complexity levels to identify potential trends. Furthermore, we evaluate the effectiveness of a privacy mechanism by controlling **Differential Privacy Defense**. This factor is binary: the defense, implemented as an additional post-processing layer rather than during training, modifies the model's final output (after the softmax stage) through a differentially private transformation. By applying an exponential mechanism that perturbs probabilities while preserving their relative ranking, the layer injects uncertainty that misleads inversion attackers yet maintains the classifier's accuracy. The central goal is to determine under which conditions and across which architectures this defense can meaningfully reduce the success of model inversion.

2.4. Model Inversion Attack and Evaluation

In this final stage, we recover representative audio samples from the trained classifiers using their gradient information. The core of our methodology is to conduct these attacks on a per-target-class basis, aiming to synthesize an input that the model confidently recognizes as belonging to a specific class.

The attack procedure is iterative. For each target class, we optimize an input signal to minimize the loss between the model's confidence score vector (i.e., the softmax output) and a one-hot encoded vector representing that target class. The attack on a given target terminates when this loss falls below a threshold of 0.005 or after a maximum of 10,000 iterations.

Across our experiments, we employ two distinct gradient-based attack strategies to generate these samples:

- **Traditional Full-Gradient Attack:** In each iteration, the gradient is computed across the entire input sample, and the whole sample is updated via backpropagation.
- **Sliding Window Attack:** This strategy updates the input in segments. It utilizes a window size and a step size to calculate the gradient for only a small portion of the input at a time, which is then updated. Extensive preliminary experiments on a standard model revealed two particularly effective parameter sets: a window size of 430 with a step size of 85 yielded the highest attack confidence, while a window size of 410 and a step size of 100 resulted in the shortest attack time. Consequently, we perform our attacks using both of these parameter configurations and average the results for a robust and comprehensive evaluation.

To rigorously evaluate the outcome of these attacks, we use three primary metrics:

- **Attack Success Rate:** An attack on a single class is deemed "successful" if the generated audio, when fed back into the target model, achieves a confidence score exceeding 80% for that class. The overall success rate is then calculated as the percentage of all target classes in a model that were successfully attacked, indicating the model's general vulnerability.

- **Attack Time:** This metric measures the total time elapsed to complete a successful attack, providing insight into the computational efficiency of the inversion process.
- **Reconstruction Quality (D-Vector Distance):** To quantitatively assess how closely the reconstructed audio resembles the original training data, we employ a separate, pre-trained generalized model. This model is trained using all audio samples from multiple datasets to serve as a universal feature extractor that generalizes across different sources and conditions. For both the reconstructed sample and the original samples of the target class, we feed them into this model and extract their respective D-Vectors—high-level feature embeddings—from its final layer. These embeddings are treated as representative feature vectors, effectively capturing the perceptual and structural characteristics of the audio. The reconstruction quality is then measured by the Euclidean distance between the D-Vector of the synthesized audio and the centroid of the D-Vectors computed from the original training samples of the corresponding class.

Within this stage, we also control one final variable: **Real-Audio Seed Initialization**. We specifically evaluate the impact of using an actual audio clip from the dataset as the initial seed for the inversion attack. This condition is compared against standard synthetic initializations (e.g., zeros, ones, constants, random noise) to determine if a more realistic starting point significantly alters the attack's trajectory and final reconstruction quality. For central tendency reporting, all results are averaged across multiple seeds and initializations.

2.5. Implementation Environment and Dataset Summary

All experiments were conducted on an Apple MacBook Pro equipped with an M2 Max chip (12-core CPU, 38-core GPU, and 32 GB unified memory). Model training and inversion procedures were implemented in Python 3.9 with Tensorflow 2.13, executed locally without external GPU clusters. The datasets used in this study include 50_Speakers [4], UrbanSound8K [5], and TIMIT [6], which together provide both speech and non-speech audio for evaluating model generalization. Table 2 summarizes the dataset composition and selection policy.

Table 2: Datasets Information

Dataset	Classes	Clips per Class (selection rule)
50_Speakers	50	1000 randomly selected per class (if < 1000, all included)
UrbanSound8K	10	1000 randomly selected per class (if < 1000, all included)
TIMIT	463	200 randomly selected per class

3. RESULTS AND DISCUSSIONS

3.1. Results

We analyze the influence of various factors, including audio clip length, model architecture, feature extraction methods, initialization strategies, model complexity, defensive measures, and dataset characteristics on attack efficacy.

1) Effect of Audio Clip Length and Normalization

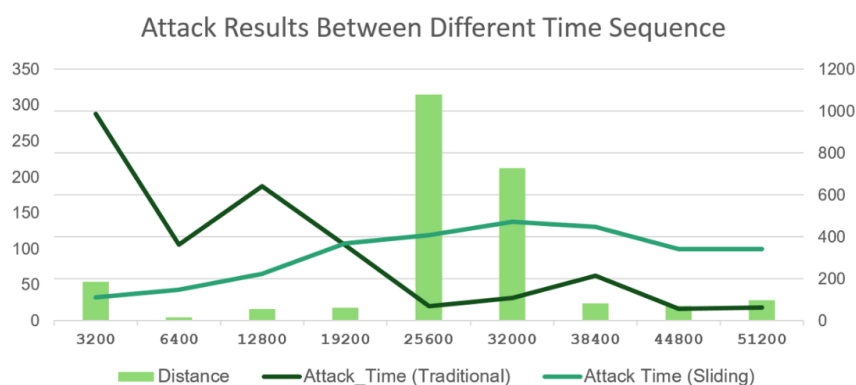


Figure 3: Effect Between Audio Length and Attack Performance

We first analyzed how the duration of an audio clip affects the computational cost of each attack method. For the sliding attack, it performs better on shorter clips using a significantly lower attack time. In contrast, the traditional attack showed a steady improvement in cost as clip length increased.

Interestingly, we observed a distinct Resistance Zone — a range in which the target model becomes particularly resistant to inversion. In our experiments, this zone appeared around 1.6 to 2 seconds (approximately 25,600–32,000 samples). Within this range, the sliding attack incurred the highest computational cost and produced reconstructions least similar to the original training samples, indicating minimal information leakage. This suggests that the model's internal feature encoding may exhibit a localized robustness effect tied to temporal structure. However, since this phenomenon may vary across datasets and architectures, further investigation is needed to generalize the existence and nature of this Resistance Zone.

2) Impact of End-to-End vs. Non-End-to-End Architectures

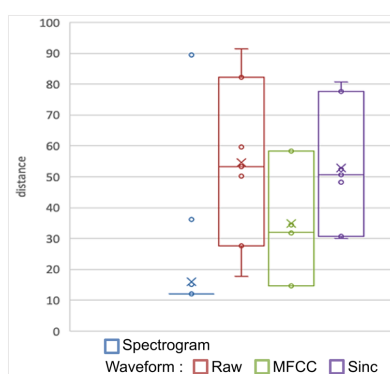


Figure 4: Spectrogram Based Model vs Waveform

The choice between end-to-end and non-end-to-end architectures significantly impacts the quality of the reconstructed audio samples. We observed that attacks on spectrogram-based models produced

results that were more perceptually similar to the original audio. As illustrated in the box plot in Figure 4, the Euclidean distances between the synthesized outputs and original samples were substantially smaller for spectrogram-based attacks (50_spec) compared to those on waveform-based models. This suggests that the richer time-frequency features inherent in spectrograms allow for the creation of more subtle and less distorted reconstructions.

3) Influence of Feature Extraction Tactics

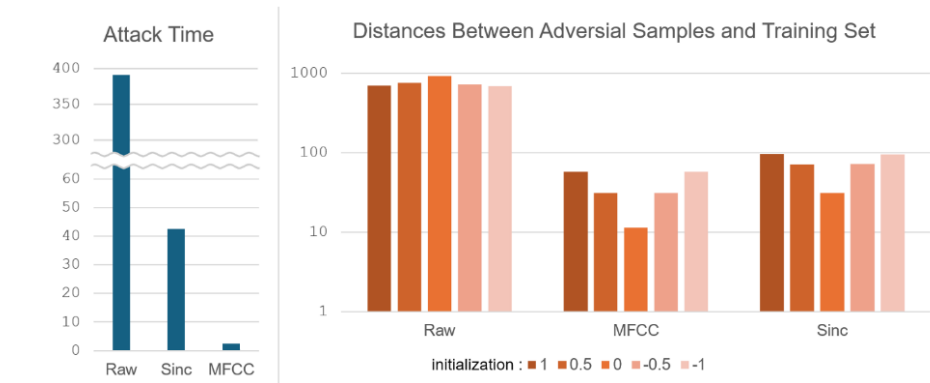


Figure 5: Comparison Between Different Feature Extraction Architecture

Table 1: Feature Size of Different Feature Extraction Architectures

	Length	Dimensions	Feature Size
Raw	3072	80	245760
MFCC	350	1	350
SincNet	200	80	16000

On end-to-end architectures, the method of feature extraction is a critical factor in determining attack outcomes. As shown in Figure 5, models that operate on raw waveforms (Raw) retain a high dimensionality of features. This characteristic leads to significantly longer attack times and results in reconstructed samples that deviate more from the original training examples. Conversely, architectures that employ early-stage feature dimensionality reduction, such as MFCC and SincNet, are more vulnerable to faster attacks. However, this speed comes at the cost of producing synthesized audio that is more closely related to the training data, as evidenced by their smaller distances.

4) Effect of Initialization Strategy

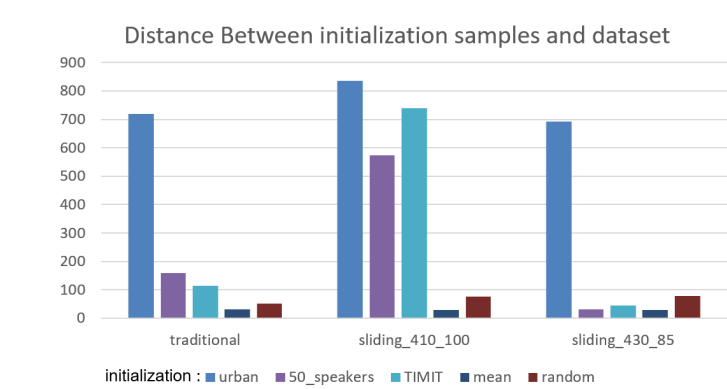


Figure 6: Comparison Between Different Initialization Samples

The initialization method used to generate adversarial samples plays a crucial role in attack performance. We investigated various initialization tactics, including using existing audio samples versus random or mean-based initialization. The results, presented in Figure 5, show that initializing an attack with a real audio sample (e.g., from the urban or TIMIT datasets) often performs worse, leading to higher attack costs than mean-based or random initialization. This suggests that the complex and inherent structures of genuine audio are more difficult to modify into adversarial examples without causing significant, easily detectable distortion.

5) Impact of Model Complexity

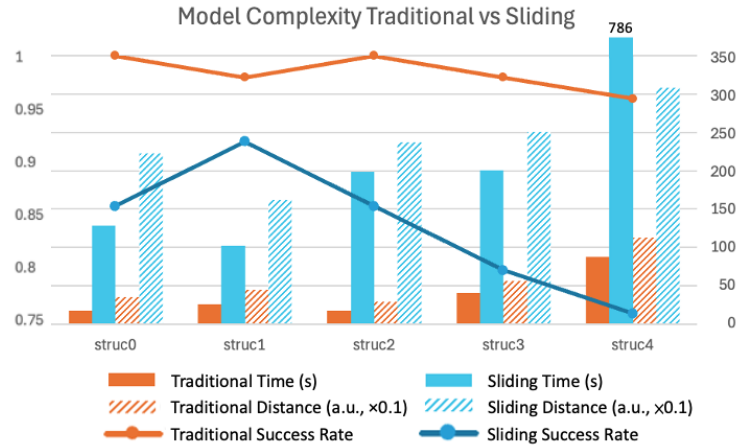


Figure 7: Performance Comparison of Traditional and Sliding Attacks across Models of Varying Complexity

We also analyzed how the structural complexity of a model affects its vulnerability. Our observations, captured in Figure 7, indicate a direct correlation between model complexity and attack difficulty. As the model depth and expressiveness increase (struc 0 through struc 4), the time required to execute a successful attack generally rises. Concurrently, key success metrics, such as prediction confidence and success rate, tend to decline. Furthermore, we observed a critical difference in how attack strategies responded to increased complexity. While the success of all attacks diminished on deeper models, the performance of the sliding attack decayed significantly faster than that of the traditional full-gradient attack.

6) Evaluation of Differential Privacy as a Defense

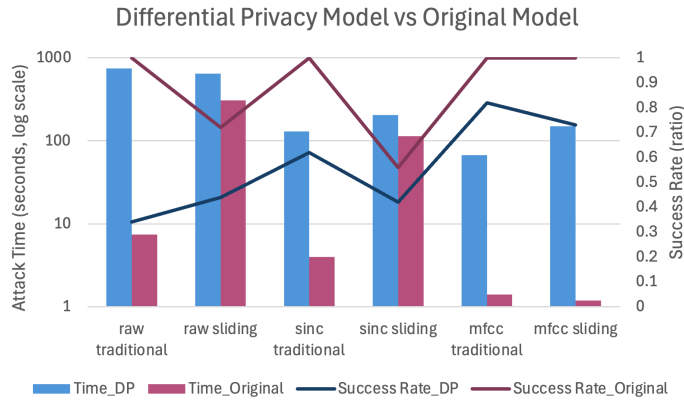


Figure 9: Effectiveness of Differential Privacy Defense across Different Model Architectures

The implementation of defensive mechanisms like differential privacy (DP) was found to be effective in hardening models against attacks. As shown in Figure 9, integrating DP into the target models significantly increased the attack cost (Time_DP vs. Time_Original) and reduced the attack success rate (Success Rate_DP vs. Success Rate_Original). This effect was particularly pronounced for Raw and SincNet models. In contrast, the performance improvement on the MFCC model was less significant, which suggests that its utility is highly dependent on the underlying model architecture.

3.2. Discussion: A Hierarchical Information Processing Framework

To understand the phenomena observed in our experiments, we propose a theoretical framework centered on the deep neural network as a Hierarchical Information Filter. In this model, the raw input audio contains the richest, most detailed, but also the most local information. As this information propagates through each layer of the network, the model learns to filter out details it deems irrelevant for the final classification task, combining the retained features into more abstract, high-level representations. Consequently, the deeper the information travels, the more global its representation becomes. This refinement process from "local" to "global" is key to understanding system vulnerabilities.

1) Attack Strategies and Information Globality

This framework clearly explains the diverging effectiveness of different attack strategies.

- **Full-Gradient Attack (Traditional Attack):** This attack optimizes against the model whole input, targeting the most global information representation. It is therefore more effective when a model's decision relies heavily on the integration of long sequences of information, which occurs in two scenarios: (1) with long audio inputs, where the model must condense extensive information into a global judgment, and (2) in complex, deep models, which are specifically designed to learn abstract, global features. The Full-Gradient attack directly targets the linchpin of the final decision, explaining why its performance decays much more slowly on complex models.
- **Sliding Attack:** The core characteristic of this attack is its locality, as it independently identifies perturbations within short time windows of the audio. This strategy exploits vulnerabilities in how the model processes early-stage, local information. It is thus more advantageous when: (1) the audio input is short, making the information inherently local, and (2) the model is simple and shallow, where the final decision still relies heavily on these unfiltered local features. As models deepen or inputs lengthen, the importance of these local vulnerabilities diminishes, limiting the sliding attack's efficacy. This finding provides a critical qualification to the work of Pizzi et al. (2023), demonstrating that the superiority of the sliding attack is not absolute but is instead a function of the model's architectural simplicity.

2) Architectural Design as an Information Filter

A model's architecture, particularly its feature extraction front-end, acts as an explicit information filter or bottleneck.

- **Raw and SincNet Architectures:** These architectures provide the model with a wide and rich information input. The Raw model passes most information through convolution, while SincNet uses learnable filters to dynamically capture features. In both cases, the model is forced to learn its own complex, high-dimensional feature extraction process. This inherent complexity poses a significant challenge for an attacker, as inverting a process that the model has learned autonomously is exceedingly difficult.

- **MFCC Architecture:** In stark contrast, MFCC imposes a fixed and extremely narrow information bottleneck at the very beginning. It uses a rigid, non-adaptive set of algorithms to discard a vast amount of information. The model's defensive capability becomes almost entirely dependent on this fragile, easily-bypassed bottleneck. Once an attacker can operate within this simplified feature space, the entire model's defense is compromised.

3) Differential Privacy: An Architecture-Dependent Defense Amplifier

Our investigation into differential privacy (DP) yields a critical insight: DP is not a universal shield but rather a "defense amplifier" whose effectiveness is highly contingent on the model's intrinsic architectural robustness. Our key contribution is to elucidate this dependency:

On Raw and SincNet models, DP demonstrates a significant defensive boost. This is because these architectures already present a high barrier to attackers—a complex, high-dimensional information space. DP's role is to further obfuscate gradient information within this already intricate maze, akin to contaminating the "gradient map" an attacker relies on. When a strong foundational defense (the complex architecture) is already in place, the gradient obfuscation from DP is highly effective at increasing attack cost and reducing success rates.

However, on the MFCC model, DP's effect is negligible. The fundamental weakness of MFCC is its oversimplified front-end information bottleneck. This bottleneck is fixed, predictable, and discards crucial information before gradients can even be computed for it. An attacker can easily craft an attack in this simplified feature domain. In this scenario, even if the back-end gradients are obfuscated by DP, it cannot compensate for the fundamental architectural flaw at the front end. This is analogous to putting a strong lock on the main gate of a castle that has no walls—attackers will simply walk around it.

Therefore, we conclude that differential privacy should be viewed as a tool to enhance robust architectures, not as a panacea for vulnerable ones. Its ability to amplify defensive capabilities is only realized when the model itself has sufficient depth and complexity to provide a meaningful baseline of security.

4. CONCLUSION

This study examines how the architectural design of speaker recognition models affects their susceptibility to model inversion attacks, and identifies under what conditions differential privacy can provide effective protection. From the attacker's perspective, our results show that defensive robustness is not incidental—it is determined by architectural depth and signal design.

In particular, robustness arises from waveform-based architectures with sufficient depth or complexity. The Raw model exhibits the highest defense capability, while SincNet represents a trade-off between training stability and defensive strength—its learnable filter design renders it slightly more vulnerable than the Raw architecture, it also improves training stability and convergence efficiency. In contrast, MFCC-based architectures impose rigid, low-dimensional bottlenecks that discard critical information early, making them more vulnerable to inversion.

We further identify a defensive "sweet spot" where both model depth and input duration jointly optimize robustness. For input sequences around 1.6-2 seconds, models maintain stable and global representations, substantially increasing attack cost.

Conversely, spectrogram-based models, which over exposes time-frequency details, are ill-suited for privacy-sensitive applications. Moreover, differential privacy functions as an architecture-dependent amplifier rather than a universal shield. In robust architectures such as Raw or SincNet, differential privacy effectively amplifies defense by injecting gradient uncertainty into already complex feature spaces. However, in weak architectures like MFCC, differential privacy provides negligible improvement since the fundamental vulnerability lies in the fixed, simplified front-end.

To interpret these results, we propose a Hierarchical Information Processing Framework, conceptualizing deep models as multi-level information filters that transform local features into

global abstractions. The success of an attack depends on how architectures preserve or compress these features across layers—linking our theoretical view with empirical evidence. Ultimately, the contribution of this work lies in providing architecture-level defense guidelines derived from the attacker’s perspective.

Our findings suggest that for privacy-resilient speaker recognition, one should adopt deep, waveform-based architectures (Raw / SincNet), maintain input lengths around 1.6–2 seconds, and treat differential privacy as a defense amplifier rather than a standalone solution. This attack-driven analysis establishes both a theoretical and practical foundation for the next generation of privacy-robust speaker recognition systems.

Nevertheless, current gradient-based attacks remain limited in perceptual realism—the generated audio, though classifiable by models, often sounds meaningless to humans. Exploring generative attack frameworks (e.g., GAN-based) that can produce semantically coherent speech will be a promising direction for future work, potentially revealing deeper insights into information leakage and enabling stronger, perception-aware defenses.

ACKNOWLEDGEMENTS

The author would like to express his deepest gratitude to Mr. Yi-Hsiung Chao and Professor Greg Lee for their invaluable guidance, insightful advice, and continuous encouragement throughout the development of this research. Their expertise in computer science and machine learning greatly inspired the design and refinement of this study.

The author also extends sincere thanks to the Department of Science and Mathematics Education, National Hualien High School, for providing research facilities and technical resources.

Finally, heartfelt appreciation is given to classmates and family members for their constant support and understanding during the experimentation and writing process.

REFERENCES

- [1] Ravanelli, M., & Bengio, Y. (2019). Speaker Recognition from Raw Waveform with SincNet. In 2018 IEEE Spoken Language Technology Workshop (SLT) (pp. 1021–1028). IEEE. <https://doi.org/10.1109/SLT.2018.8639585>
- [2] Pizzi, K., Boenisch, F., Sahin, U., Böttinger, K. (2022) Introducing model inversion attacks on automatic speaker recognition. Proc. 2nd Symposium on Security and Privacy in Speech Communication, 11-16, doi: 10.21437/SPSC.2022-3
- [3] Ye, D., Shen, S., Zhu, T., Liu, B., & Zhou, W. (2022). One parameter defense: Defending against data inference attacks via differential privacy. IEEE Transactions on Information Forensics and Security, 17, 1466–1480. <https://doi.org/10.1109/TIFS.2022.3163591>
- [4] Jain, V. (2019). Speaker recognition audio dataset. Kaggle. <https://www.kaggle.com/datasets/vjcalling/speaker-recognition-audio-dataset>
- [5] Salamon, J., Jacoby, C., & Bello, J. P. (2014, November). UrbanSound8K dataset. <https://urbansounddataset.weebly.com/urbansound8k.html>
- [6] Fekadu, M. (2019). DARPA TIMIT acoustic-phonetic continuous speech. Kaggle. <https://www.kaggle.com/datasets/mfekadu/darpa-timit-acousticphonetic-continuous-speech>

BIOGRAPHIES OF AUTHORS

	Xuan-Yu Lin is a student researcher at National Hualien Senior High School, Taiwan, with a strong interest in artificial intelligence and computer science. His current research focuses on model inversion attacks and privacy-preserving architectures for speaker recognition systems. He is also passionate about natural language processing, deep learning, and information security, and aims to pursue further studies in these interdisciplinary areas of AI and electrical engineering.
	Chia-Wei Chuang is a student at National Hualien Senior High School, Taiwan. His current research focuses on model inversion attacks and defense architectures for speaker recognition systems. His broader interests include language processing, machine learning, and related engineering fields. He has also participated in the development of multimodal AI communication systems designed to assist patients with aphasia. He plans to continue his academic journey in artificial intelligence and electrical engineering.